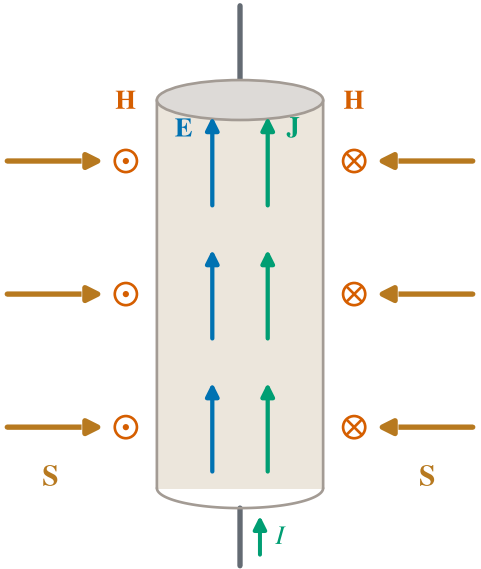
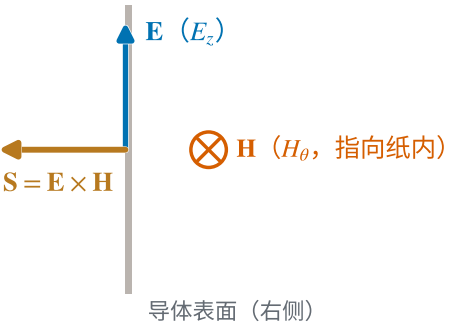


单位长度流入：$2\pi a\, E_z H_\theta = E_z I = I^2 R/\ell$

$$H_\theta = \frac{I}{2\pi a}$$（安培定律）



流入的功率恰好等于导体内的欧姆热